Seat No.

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 1 **Year:** 2019
Subject: 392153 Chemistry III **Section:** 15-18
Date: 15 August 2019 **Time:** 13.00-15.00

Name:______________________ID:____________________Class:__________

Directions: The test is designed to measure your comprehension. The test is divided into 2 sections. There will be 13 pages (including this page), 7 questions and they are worth 80 marks (40 points).

- Answer the questions on this test papers.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can't be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

INSTRUCTIONS: Written Response

Students will be expected to communicate your knowledge and understanding of chemical principles in a clear and logical manner. Students' steps and assumptions leading to a solution must be written in the spaces below the questions. Answers must include units where appropriate and be given to the correct number of significant figures. **For questions involving calculation and explanation, full marks will NOT be given for providing only an answer.**

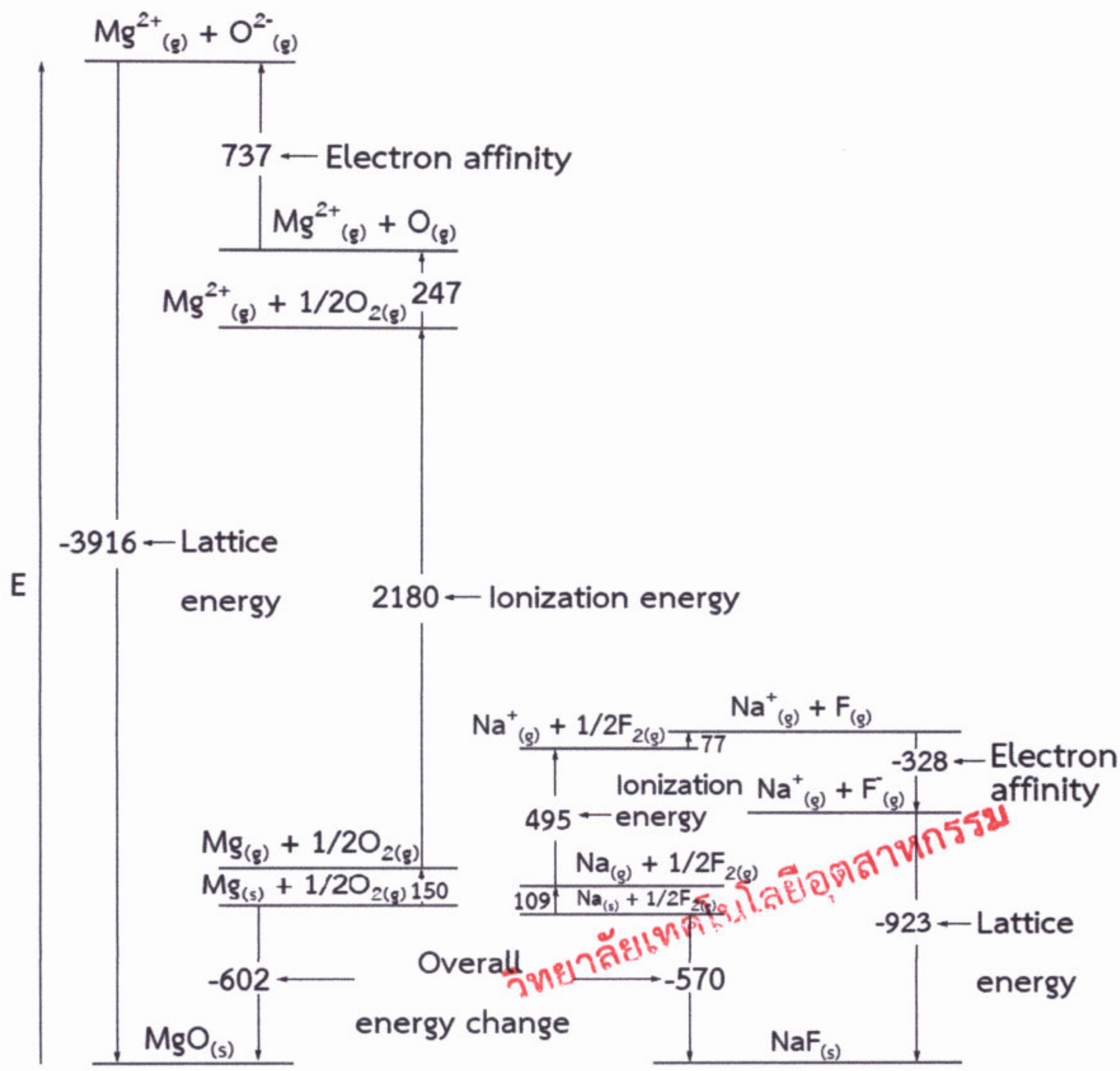
Part 1: Total score 40 marks = 20%

1 . VSEPR or Valence Shell Electron Pair Repulsion theory is the theory applied to determine the shape of molecules. (10 marks)

1.1 Please give four assumptions of this theory to consider the shape of molecules?

1.2 How does the shape of molecules correspond to dipole moment of the molecules?

2. Use the Lattice energy flow chart below to discuss about the comparison of the energy changes involved in the formation of solid sodium fluoride and solid magnesium oxide. (10 marks)



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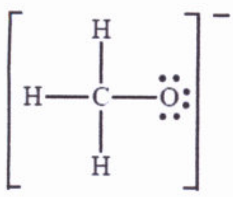
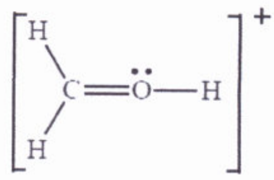
4. Answer the following questions about formal charges.

4.1 What is the relationship between having full valence shells and formal charges? Do full valence shells always result in a formal charge of zero?

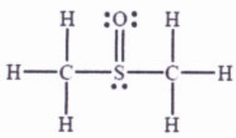
4.2 Calculate the formal charge on each atom in the following Lewis structures.



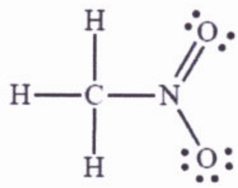
Methoxide ion



An oxonium ion



Dimethyl sulfoxide



Nitromethane

Asst.Prof.Dr. Rungsima Yeetsorn

<u>Part 2</u>: Total score 40 marks = 20%

1. Consider this reaction: $CH_3OH_{(l)} + O_{2(g)} \longrightarrow H_2O_{(l)} + CO_{2(g)}$

1.1) Please balance the chemical reaction. (2 marks)

1.2) What is the value of ΔH of the following reaction? (Hint: using bond energies in the table below) (10 marks)

1.3) What kind of the reaction (endothermic or exothermic reaction)? And, explain your reason. (3 marks)

1.4) If you use 4 moles of CH_3OH of the chemical reaction, please calculate the value of ΔH. (5 marks)

Single Bonds						Multiple Bonds	
H—H	432	N—H	391	I—I	149	C=C	614
H—F	565	N—N	160	I—Cl	208	C≡C	839
H—Cl	427	N—F	272	I—Br	175	O=O	495
H—Br	363	N—Cl	200			C=O*	745
H—I	295	N—Br	243	S—H	347	C≡O	1072
		N—O	201	S—F	327	N=O	607
C—H	413	O—H	467	S—Cl	253	N=N	418
C—C	347	O—O	146	S—Br	218	N≡N	941
C—N	305	O—F	190	S—S	266	C≡N	891
C—O	358	O—Cl	203			C=N	615
C—F	485	O—I	234	Si—Si	340		
C—Cl	339			Si—H	393		
C—Br	276	F—F	154	Si—C	360		
C—I	240	F—Cl	253	Si—O	452		
C—S	259	F—Br	237				
		Cl—Cl	239				
		Cl—Br	218				
		Br—Br	193				

*C=O(CO_2) = 799

2. Answer the following questions about intermolecular forces.

2.1) Please identify types of intermolecular forces (London force, Dipole-dipole force, Hydrogen bond) in the substances below. (10 marks)

Substance	Intermolecular force
HF	
CH_2F_2	
NH_3	
CO_2	
Br_2	

2.2) Arrange the following compounds (HF, CH_2F_2, NH_3, CO_2, Br_2) from the highest to the lowest boiling point. To achieve full marks, please explain your reasons. (5 marks)

3. What is the meaning of "like dissolved like" in solution? Please give three examples of the "like dissolved like". (10 marks)

PERIODIC TABLE OF ELEMENTS

Atomic #
Symbol
Name
Weight

C Solid
Hg Liquid
H Gas
Rf Unknown

Metals: Alkali metals | Alkaline earth metals | Lanthanoids (Lanthanides) | Actinoids (Actinides) | Transition metals | Post-transition metals

Metalloids

Nonmetals: Other nonmetals | Noble gases

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15 (Pnictogens)	16 (Chalcogens)	17 (Halogens)	18
1	1 **H** Hydrogen 1.008																	2 **He** Helium 4.0026
2	3 **Li** Lithium 6.94	4 **Be** Beryllium 9.0122											5 **B** Boron 10.81	6 **C** Carbon 12.011	7 **N** Nitrogen 14.007	8 **O** Oxygen 15.999	9 **F** Fluorine 18.998	10 **Ne** Neon 20.180
3	11 **Na** Sodium 22.990	12 **Mg** Magnesium 24.305											13 **Al** Aluminium 26.982	14 **Si** Silicon 28.085	15 **P** Phosphorus 30.974	16 **S** Sulfur 32.06	17 **Cl** Chlorine 35.45	18 **Ar** Argon 39.948
4	19 **K** Potassium 39.098	20 **Ca** Calcium 40.078	21 **Sc** Scandium 44.956	22 **Ti** Titanium 47.867	23 **V** Vanadium 50.942	24 **Cr** Chromium 51.996	25 **Mn** Manganese 54.938	26 **Fe** Iron 55.845	27 **Co** Cobalt 58.933	28 **Ni** Nickel 58.693	29 **Cu** Copper 63.546	30 **Zn** Zinc 65.38	31 **Ga** Gallium 69.723	32 **Ge** Germanium 72.630	33 **As** Arsenic 74.922	34 **Se** Selenium 78.971	35 **Br** Bromine 79.904	36 **Kr** Krypton 83.798
5	37 **Rb** Rubidium 85.468	38 **Sr** Strontium 87.62	39 **Y** Yttrium 88.906	40 **Zr** Zirconium 91.224	41 **Nb** Niobium 92.906	42 **Mo** Molybdenum 95.95	43 **Tc** Technetium (98)	44 **Ru** Ruthenium 101.07	45 **Rh** Rhodium 102.91	46 **Pd** Palladium 106.42	47 **Ag** Silver 107.87	48 **Cd** Cadmium 112.41	49 **In** Indium 114.82	50 **Sn** Tin 118.71	51 **Sb** Antimony 121.76	52 **Te** Tellurium 127.60	53 **I** Iodine 126.90	54 **Xe** Xenon 131.29
6	55 **Cs** Caesium 132.91	56 **Ba** Barium 137.33	57–71	72 **Hf** Hafnium 178.49	73 **Ta** Tantalum 180.95	74 **W** Tungsten 183.84	75 **Re** Rhenium 186.21	76 **Os** Osmium 190.23	77 **Ir** Iridium 192.22	78 **Pt** Platinum 195.08	79 **Au** Gold 196.97	80 **Hg** Mercury 200.59	81 **Tl** Thallium 204.38	82 **Pb** Lead 207.2	83 **Bi** Bismuth 208.98	84 **Po** Polonium (209)	85 **At** Astatine (210)	86 **Rn** Radon (222)
7	87 **Fr** Francium (223)	88 **Ra** Radium (226)	89–103	104 **Rf** Rutherfordium (267)	105 **Db** Dubnium (268)	106 **Sg** Seaborgium (269)	107 **Bh** Bohrium (270)	108 **Hs** Hassium (277)	109 **Mt** Meitnerium (278)	110 **Ds** Darmstadtium (281)	111 **Rg** Roentgenium (282)	112 **Cn** Copernicium (285)	113 **Nh** Nihonium (286)	114 **Fl** Flerovium (289)	115 **Mc** Moscovium (290)	116 **Lv** Livermorium (293)	117 **Ts** Tennessine (294)	118 **Og** Oganesson (294)

For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.

6	57 **La** Lanthanum 138.91	58 **Ce** Cerium 140.12	59 **Pr** Praseodymium 140.91	60 **Nd** Neodymium 144.24	61 **Pm** Promethium (145)	62 **Sm** Samarium 150.36	63 **Eu** Europium 151.96	64 **Gd** Gadolinium 157.25	65 **Tb** Terbium 158.93	66 **Dy** Dysprosium 162.50	67 **Ho** Holmium 164.93	68 **Er** Erbium 167.26	69 **Tm** Thulium 168.93	70 **Yb** Ytterbium 173.05	71 **Lu** Lutetium 174.97
7	89 **Ac** Actinium (227)	90 **Th** Thorium 232.04	91 **Pa** Protactinium 231.04	92 **U** Uranium 238.03	93 **Np** Neptunium (237)	94 **Pu** Plutonium (244)	95 **Am** Americium (243)	96 **Cm** Curium (247)	97 **Bk** Berkelium (247)	98 **Cf** Californium (251)	99 **Es** Einsteinium (252)	100 **Fm** Fermium (257)	101 **Md** Mendelevium (258)	102 **No** Nobelium (259)	103 **Lr** Lawrencium (266)

Carbon
12.011

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Electronegativity values

H 2.1																	
Li 1.0	Be 1.5											B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	
Na 0.9	Mg 1.2											Al 1.5	Si 1.8	P 2.2	S 2.5	Cl 3.0	
K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.8	Ni 1.8	Cu 1.9	Zn 1.6	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8	
Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5	
Cs 0.7	Ba 0.9	La-Lu 1.1-1.2	Hf 1.3	Ta 1.5	W 1.7	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Tl 1.8	Pb 1.8	Bi 1.9	Po 2.0	At 2.2	
Fr 0.7	Ra 0.9	Ac-No 1.1-1.7															

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